

FINANCE FOR COMPUTER SCIENCE (FIN-3103)

Practice Assessment - Semester 4

Total Marks: 50

Instructions:

- Attempt all questions.
- Marks for each question are indicated on the right side.
- Show clear calculations where necessary.

SECTION A: CONCEPTUAL SHORT QUESTIONS (20 Marks)

Q1. Explain the formula for calculating the Yield to Maturity (YTM) of a bond and briefly describe what YTM actually represents for an investor. *(2 Marks)*

Q2. Differentiate between Common Stock and Preferred Stock in terms of dividend payments, voting rights, and priority during liquidation. *(2 Marks)*

Q3. What is the Net Present Value (NPV) method in capital budgeting? What decision rule does a financial manager follow when using NPV to accept or reject a project? *(2 Marks)*

Q4. State the primary purpose of financial forecasting for a business. Mention one key limitation of relying on historical financial data for these forecasts. *(2 Marks)*

Q5. Define the Internal Rate of Return (IRR) method in capital budgeting. How is the IRR used to make an investment decision? *(2 Marks)*

Q6. Define the Cash Conversion Cycle (CCC). Write down the three individual periods that are used to calculate the CCC. *(2 Marks)*

Q7. Explain the difference between Current Assets and Current Liabilities. Give one example of each. *(2 Marks)*

Q8. What is meant by "Working Capital"? How does a company's working capital management affect its daily operations? *(2 Marks)*

Q9. In the context of capital budgeting, briefly explain "Project Classification" and "Liquidation Value." *(2 Marks)*

Q10. Explain the inverse relationship between bond prices and market interest rates. If interest rates go down, what happens to the price of an existing bond? *(2 Marks)*

SECTION B: ANALYTICAL & CALCULATION QUESTIONS (20 Marks)

Q11. Cash Conversion Cycle Calculation (10 Marks)

XYZ Corporation, a manufacturing firm, reported the following financial details for the fiscal year 2025:

- Annual Sales: \$4,500,000
 - Cost of Goods Sold (COGS): \$3,200,000
 - Average Inventory: \$600,000
 - Average Accounts Receivable: \$750,000
 - Average Accounts Payable: \$400,000
- (Note: Assume a 365-day year for your calculations.)

Required:

- a) Calculate the Inventory Conversion Period (ICP). (2 Marks)
- b) Calculate the Receivables Conversion Period (RCP). (2 Marks)
- c) Calculate the Payables Deferral Period (PDP). (2 Marks)
- d) Determine the total Cash Conversion Cycle (CCC) for the firm. (2 Marks)
- e) Based on your result, interpret whether the company's cash is efficiently tied up in operations. Suggest one specific operational strategy the firm could use to shorten this cycle. (2 Marks)

Q12. Payback Period Analysis (10 Marks)

A company is evaluating two independent investment opportunities, Project M and Project N.

- **Project M:** Requires an initial investment of **\$120,000****. It is expected to generate a constant annual cash inflow of ****\$24,000** for the next 8 years.
- **Project N:** Requires an initial investment of **\$180,000**. The estimated cash inflows for the next 5 years are as follows:
 - Year 1: \$30,000
 - Year 2: \$50,000
 - Year 3: \$60,000
 - Year 4: \$70,000
 - Year 5: \$40,000

Required:

- a) Calculate the Payback Period for Project M. *(2 Marks)*
- b) Calculate the Payback Period for Project N. *(4 Marks)*
- c) If the company's maximum permissible payback period is **4 years**, which project(s) should be accepted based strictly on this criterion? *(2 Marks)*
- d) Identify two major limitations of using the Payback Period method for making significant investment decisions. *(2 Marks)*

SECTION C: APPLICATION OF VALUATION CONCEPTS (10 Marks)

Q13. Stock and Bond Valuation *(10 Marks)*

- a) **Stock Valuation (Gordon Growth Model):** AlphaTech Inc. has just paid an annual dividend of **\$2.00** per share. The company's dividends are expected to grow at a constant annual rate of **5%** indefinitely. If investors require a **12%** rate of return on this type of stock, what is the intrinsic value of AlphaTech's stock today? *(5 Marks)*
- b) **Bond Price Dynamics:** A company issues a bond with a face value of \$1,000 and an annual coupon rate of 7%. The current market Yield to Maturity (YTM) for similar bonds is 9%.
 - i) Will this bond trade at a **premium**, a **discount**, or at **par**? Explain your reasoning. *(2 Marks)*
 - ii) If the market YTM suddenly drops to 5%, briefly explain what will happen to the market price of this bond. *(3 Marks)*